

Research Report on Non-Standard Architectural Precedents: Spatial Phenomenology, Tectonic Whimsy, and Radical Care

Brief Framing of My Position

The architectural position under examination asserts that beauty is a fundamental, non-negotiable value in spatial design rather than a decorative afterthought or a quality that requires secondary logical validation. It prioritises beauty as the initial point of somatic attraction, with structural integrity, spatial intelligence, and authorial craft establishing deeper respect thereafter. The body is valued as highly as the eye; space is never neutral, and elements such as light, sound, texture, threshold, exposure, and sequence affect human experience before they can be rationally articulated. This philosophy embraces productive contradictions—such as simplicity and complexity, or the mundane and the ethereal—asserting that feeling is a core component of lived utility. Finally, grounded in radical empathy, it argues that care must be spatial, material, and institutional, designing places of arrival, waiting, retreat, and transition that respect the complexities of real human beings.

Junya Ishigami as Calibration

The design philosophy of Junya Ishigami serves as an excellent calibration reference for this position. His works consistently explore the boundaries of lightness, atmosphere, and ambiguity, creating fragile spatial conditions that blur the distinction between object, landscape, body, and perception. In projects such as the House & Restaurant in Yamaguchi, Ishigami proves that beauty is not decorative but structural, using a flowing, cave-like form to evoke an ancient, weathered, and deeply emotional atmosphere.

However, when utilising Ishigami as a guide, several aspects must not be overgeneralised. His work is often misread as effortless or hyper-minimalist, yet it relies on extreme structural rigour, manual labour, and precise computational tools—such as 3D scans—to fit functional glass enclosures into organic concrete openings. Therefore, he calibrates the position by showing that ethereal beauty requires rigorous technical control rather than a abandonment of functional reality. He best calibrates the categories of subtly surreal dreamscapes and beauty through process, demonstrating that "form follows feeling" can be successfully realised through experimental, highly demanding construction methods.

Research Strategy

The research strategy adopted for this report rejects any singular, reductive reading of the architectural position. It investigates a diverse array of global precedents across multiple historical eras, scales, geographies, and typologies to provide a rich matrix of design options.

By searching for architectures that embody a delicate strangeness without falling into theatrical gimmickry, the strategy covers sacred, civic, domestic, landscape, and installation-based designs. These precedents show how physical materials and spatial sequences can engage the body, support emotional safety, and express structural playfulness.

Previously Covered Material and Continued Relevance

The following historical and theoretical materials from previous research passes have been evaluated to establish their continued relevance to this expanded study.

Previous Case Studies

- Z33 House for Contemporary Art by Francesca Torzo, completed in 2019, remains relevant to the revised research because its hand-laid brick patterns and subtle shifts in domestic scale illustrate the choreography of quiet, subtly surreal dreamscapes.
- The Green Corner Building by Studio Anne Holtrop, completed in 2020, remains relevant because its sand-cast structural components illustrate how raw, process-driven form-finding can challenge typical geometric perfection.
- Walmer Yard by Peter Salter, completed in 2016, remains relevant because its dense material textures, heavy shadows, and deep thresholds offer a premier example of raw, physical refuge.
- Chuzhi by Wallmakers, completed in 2022, remains relevant because its swirling, debris-filled subterranean vaults show how experimental, eco-responsive form-finding can protect the sensory dignity of the occupant.
- Hirosaki Museum of Contemporary Art by Atelier Tsuyoshi Tane Architects, completed in 2020, remains relevant because the preservation of its historical brickwork and the introduction of a reflective metal roof show how architecture holds the contradiction of memory and future.

Previous Theories and Trends

- Atmospheric Aesthetics and the "Ecstasies of Things" by Gernot Böhme remains relevant because it theorises how raw materials radiate their sensory qualities to actively generate spatial atmospheres.
- Embodied Phenomenology and Sensory Vision by Maurice Merleau-Ponty and Juhani Pallasmaa remains relevant because it explains how spatial sequences affect human emotion before intellectual rationalisation occurs.
- Post-Digital Embodied Computation and Anexact Architecture remains relevant because it demonstrates that digital design tools can capture and translate physical material variations without imposing slick, generic perfection.

Interpretive Categories for Case Studies

To structure the precedent analysis, eight distinct interpretive categories are established, reflecting the nuanced layers of the architectural position.

The first category, subtly surreal dreamscapes, focuses on spaces that cultivate controlled ambiguity, unreality, and delicate strangeness to prompt heightened sensory awareness. The

second category, tectonic whimsy, examines structures where structural details, material logic, and assembly methods produce a sense of playfulness, weightlessness, or poetic delight. Under emotions, elements, and the ethereal, the research highlights architectures shaped primarily by natural forces such as light, shadow, wind, water, and material weathering. The fourth category, embodied atmosphere, looks at how sequence, density, acoustics, and temperature register on the human body before the intellect processes them. The fifth category, radical care and refuge, addresses spaces designed to support psychological safety, dignity, and sensory regulation, particularly for vulnerable populations. Liminal ritual spaces examine the architecture of transition, memory, and the thresholds between ordinary and extraordinary experience. The seventh category, beauty through process, highlights works where aesthetic value is not a preconceived image but emerges directly from experimental construction techniques and hands-on material testing. Finally, computational or experimental form-finding explores the integration of technology to capture organic irregularities, proving that computation can support sensory intuition rather than sterile complexity.

Broad Longlist of Possible Case Studies

The following longlist provides a wide range of precedents aligned with the developed architectural position. To remain accessible for auditory analysis, no tables are used, and each entry is written in clear, complete sentences.

Case study name: House & Restaurant

Architect or practice: The project was completed by Junya Ishigami and Associates in Ube, Yamaguchi Prefecture, Japan, in 2022.

Why it may fit: This domestic and culinary retreat perfectly fits the established position by presenting a cave-like sanctuary that holds the contradiction between natural soil and highly precise, modern glass enclosures. Situated in the category of subtly surreal dreamscapes, this building is a semi-obvious precedent that prioritises the sensory and emotional reactions of the body over clean, logical lines.

Form-finding strategy: The form-finding strategy involved pouring reinforced concrete directly into deep excavations in the earth, using the natural soil as a textured mould before digging out the surrounding ground to expose the raw structure. This approach achieves high atmospheric and tectonic relevance by allowing soil geology and manual labour to define the final spatial sequence.

Possible visuals: The study would benefit from 3D scans of the concrete-to-glass connections, soil excavation site diagrams, and photographs of the subterranean dining areas.

Source leads: Credible documentation can be found in detailed project descriptions in Space Magazine and official monographs from Junya Ishigami and Associates.

Case study name: Ca'n Terra

Architect or practice: The project was completed by Ensamble Studio in Menorca, Spain, in 2021.

Why it may fit: This residential retreat represents an extraordinary example of beauty through process, converting an abandoned stone quarry into an inhabitable, contemplative home. This semi-obvious project aligns with the position by showing how architecture can step back, allowing raw material and natural light to suggest the spatial layout.

Form-finding strategy: The form-finding strategy utilized advanced 3D laser scanning to record millions of coordinates inside the dark cavern, allowing the architects to interpret the geological structure and make precise, minimal cuts to let in light and air. This highlights an atmospheric strategy where digital technology is used to structure raw intuition and respect the pre-existing environment.

Possible visuals: Point cloud scans illustrating the cavern interior, architectural plans overlaid on quarry topography, and photographs of light reflecting off the stone walls are recommended.

Source leads: Useful details are available in exhibition materials from the London Design Museum and articles in international design journals like *This is paper*.

Case study name: La Trufa (The Truffle)

Architect or practice: The project was completed by Ensamble Studio on the Costa da Morte, Spain, in 2010.

Why it may fit: This small beach house is a deep cut in process-driven beauty, emulating mineral formation to create an amorphous concrete boulder that blends into the wild landscape. It supports the idea of "form follows feeling" by creating a compact, highly poetic sanctuary that frames the Atlantic horizon.

Form-finding strategy: The form-finding strategy involved digging a pit in the ground, packing the center with hay bales to represent the internal air volume, and pouring mass concrete into the interstitial spaces. Once cured, the earth was removed, and a calf named Paulina was brought in to eat the compressed hay over a year, gradually revealing the unpolished, textured interior.

Possible visuals: Process diagrams of the concrete-and-hay pouring sequence, photographs of the calf inside the space, and sections showing the ocean-facing window are highly useful.

Source leads: Refer to Blaine Brownell and Marc Swackhamer's book, *Hypernatural: Architecture's New Relationship with Nature*, and official Ensamble Studio project texts.

Case study name: PC Caritas

Architect or practice: The project was completed by Architecten de Vylder Vinck Taillieu in Melle, Belgium, in 2016.

Why it may fit: This project fits the category of radical care and refuge, transforming a partially demolished nineteenth-century psychiatric villa into an open-air public loggia and therapeutic greenhouse. This semi-obvious project aligns with the position by holding the contradictions of decay, safety, and institutional care, creating a space of dignity for patients.

Form-finding strategy: The form-finding strategy relied on direct physical engagement with the found building ruins, inserting modern glass greenhouses into the open shell and having craftsmen repair broken masonry with concrete blocks on-site without pre-determined blueprints. This tectonic approach values the existing structure as material, producing a highly emotional, open-ended spatial sequence.

Possible visuals: Photographs of the concrete block repairs, drawings showing the placement of the greenhouses, and plans of the open loggia floors are essential.

Source leads: Consult reviews in *Domus*, project portfolios by Filip Dujardin, and publications from the Flanders Architecture Institute.

Case study name: Water Cathedral

Architect or practice: The project was designed by GUN Architects and constructed in Santiago, Chile, in 2011.

Why it may fit: This temporary public pavilion belongs to the category of elemental and ethereal architecture, creating a damp, microclimatic cavern in a hot urban context. It is a deep cut that

explores water as prime atmospheric matter, slowing down the perception of time to engage the visitor's senses.

Form-finding strategy: The form-finding strategy utilized a minimal steel grid supporting a field of suspended, textile-covered pyramidal prisms filled with a granular stone substrate that regulates and slows water dripping through capillary action. This tectonic approach creates concrete stalagmites on the ground that serve as seating and water storage, offering a playful, cooling public space.

Possible visuals: Detailed drawings of the suspended textile prisms, hydraulic irrigation system diagrams, and photographs of visitors interacting with the water are recommended.

Source leads: Review the project descriptions on MoMA PS1's Young Architects Program archive and articles in international journals like *Detail*.

Case study name: Takasugi-an (The Too-High Teahouse)

Architect or practice: The project was designed by Terunobu Fujimori in Chino, Nagano Prefecture, Japan, in 2004.

Why it may fit: This compact, elevated teahouse is a semi-obvious masterpiece of tectonic whimsy, perched precariously on two chestnut trees to evoke a dreamlike, folklore-inspired atmosphere. It directly aligns with the position by treating the small building as a physical extension of the body and manipulating scale to offer a serene refuge.

Form-finding strategy: The form-finding strategy rejected professional, commercial construction methods, with the architect personally designing and hand-building the structure using raw chestnut trunks cut from a nearby mountain. Inside, the plaster and bamboo lining focuses the occupant's senses on the framed view of the surrounding valley, celebrating material honesty.

Possible visuals: Elevation drawings showing the tree-stilt supports, interior photos of the plaster-lined room, and plans of the ladder access are recommended.

Source leads: Refer to essays by Kengo Kuma on Fujimori's Jinchokan Moriya Historical Museum and profiles in international design journals.

Case study name: Lamune Onsen (Soda Pop Spa)

Architect or practice: The project was completed by Terunobu Fujimori in Takeda, Oita Prefecture, Japan, in 2005.

Why it may fit: This carbonated hot spring facility is a deep cut in the category of embodied atmosphere, using hand-crafted textures to celebrate the sensory experience of bathing. Its striking black-and-white striped facade and living pine trees sprouting from the roof towers create an otherworldly, charming environment that directly engages the bather's body.

Form-finding strategy: The form-finding strategy involved cladding the reinforced concrete frame in alternating vertical bands of carbonised cedar planks and white mortar, with hand-laminated copper roof elements capped with live vegetation. This tectonic approach blends ancient Japanese temple architecture with expressive, playful material choices to stimulate multisensorial perception.

Possible visuals: Photographs of the carbonised cedar cladding, sections through the steam-release towers, and interior photos of the plaster-lined cave-like entrances are valuable.

Source leads: Review the 2006 Venice Biennale Japanese Pavilion catalogue and essays by Maria Antonietta Crippa.

Case study name: Les Cols Pavilions and Marquee

Architect or practice: The project was completed by RCR Arquitectes in Olot, Spain, in 2011.

Why it may fit: This outdoor event and dining space belongs to the category of elemental and

ethereal architecture, subverting conventional hospitality design to evoke the feeling of family picnics in a volcanic valley. This semi-obvious project aligns with the position by using transparent polymer walls and clear acrylic furniture to dissolve boundaries, allowing light and nature to dominate.

Form-finding strategy: The form-finding strategy involved excavating the site directly into the volcanic landscape, utilizing dry-stone volcanic walls for lateral support and a lightweight polymer roof slung across steel tubes to form a translucent canopy. This tectonic approach creates a sophisticated balance of rough-raw and smooth-detailed materials, producing an ephemeral, hazy light.

Possible visuals: Site plans showing the orthogonal layout cut into volcanic earth, detailed sections of the slung polymer roof, and photos of the clear acrylic furniture are useful.

Source leads: Consult the 2017 Pritzker Prize media kit and essays in *El Croquis* focusing on RCR's early Olot works.

Case study name: Flower Kiosk at Malmö Eastern Cemetery

Architect or practice: The project was completed by Sigurd Lewerentz in Malmö, Sweden, in 1969.

Why it may fit: This tiny, raw building sits at the cemetery entrance, representing a deep cut in the category of liminal ritual spaces that mediates the transition between the ordinary world and the space of mourning. It aligns with the position by celebrating structural imperfection and accepting aging, showing how a modest retail kiosk can carry immense emotional dignity.

Form-finding strategy: The form-finding strategy utilized rough, dark brick laid with wide, uneven mortar joints and a steep, dramatic copper canopy. Its windows are entirely frameless, clipped directly to the exterior masonry to challenge typical tectonic junctions, while interior electrical wires are laid in organic patterns reminiscent of flower petals.

Possible visuals: Detailed drawings of the surface-mounted glass clips, interior electrical wiring layout diagrams, and photographs of the weathered brick patina are recommended.

Source leads: Read Janne Ahlin's monograph, *Sigurd Lewerentz, Architect: 1885–1975*, and essays in *Drawing Matter*.

Case study name: Sancaklar Mosque

Architect or practice: The project was completed by Emre Arolat Architects in Büyükçekmece, Istanbul, Turkey, in 2012.

Why it may fit: This subterranean mosque belongs to the category of liminal ritual spaces, rejecting traditional Ottoman domes and decorative clutter to focus entirely on a cave-like, silent prayer hall. This semi-obvious project aligns with the position by using natural light as its primary ornament, creating a dramatic, non-rectilinear space that prioritises the physical presence of the worshipper.

Form-finding strategy: The form-finding strategy set the building into a natural slope, terracing it with rough local slate steps that merge into the prairie landscape, leaving only a stone canopy and minaret visible above ground. This approach creates a dramatic transition that enhances the spiritual experience, allowing the interior exposed concrete to strengthen the feeling of purification.

Possible visuals: Longitudinal sections showing the subterranean excavation, site plans illustrating the terraced landscape, and photographs of light leaking through the fractured Qiblah wall are highly recommended.

Source leads: Read "Sancaklar Mosque: Displacing the Familiar" and project essays published in international architectural journals.

Case study name: Teletón Children's Rehabilitation Center

Architect or practice: The project was completed by Gabinete de Arquitectura in Lambaré, Paraguay, in 2013.

Why it may fit: This project fits the category of radical care and refuge, providing physical rehabilitation within a highly expressive brick landscape that challenges typical sterile clinic typologies. It is a deep cut that values structural integrity and material warmth as therapeutic assets, proving that social care must be matched by spatial and tectonic dignity.

Form-finding strategy: The form-finding strategy involved prefabricating structural brick-and-mortar panels on the ground using low-cost, local materials, tilting them upward to form vaults, arches, and self-supporting screen walls that shade the interior. This tectonic approach turns cheap, discarded bricks into a highly playful and expressive canopy that protects the sensory dignity of the patients.

Possible visuals: Photographs of the brick prefabricated panels laid on the ground, structural diagrams of the vaults, and photographs of patients moving through the shaded brick arcades are essential.

Source leads: Refer to Nick Gelpi's book, *The Architecture of Full-Scale Mock-Ups*, and interviews with Solano Benítez.

Case study name: Gando Primary School

Architect or practice: The project was completed by Francis Kéré in Gando, Burkina Faso, in 2001.

Why it may fit: This school fits the category of radical care and refuge, designed to provide a comfortable, naturally ventilated learning environment in an extreme desert climate. This semi-obvious project aligns with the position by showing that high-quality, beautiful architecture can be achieved under severe economic constraints through local participation and vernacular intelligence.

Form-finding strategy: The form-finding strategy utilized hand-pressed clay-and-cement blocks for the walls, capped by a dry-stacked clay brick ceiling and an overhanging corrugated metal roof suspended on delicate steel rebar trusses to maximize passive cooling. This tectonic approach separates the roof from the classroom space, allowing hot air to escape through ceiling perforations while protecting the clay walls from rain.

Possible visuals: Detailed sections illustrating the passive cooling air currents, photographs of the hand-pressed brick construction process, and drawings of the suspended metal roof trusses are highly useful.

Source leads: Read Kéré Architecture's project profiles and the technical evaluations compiled by the Aga Khan Award jury.

Case study name: SESC Pompéia Architect or practice: The project was designed by Lina Bo Bardi and completed in São Paulo, Brazil, in 1986.

Why it may fit: This leisure center belongs to the category of radical care and refuge, converting a derelict factory into a vibrant community hub that prioritises play, communal dining, and community theatre. This famous project aligns with the position by showing that architectural care is material and social, using unpolished, raw materials to support the spontaneous activities of the working class.

Form-finding strategy: The form-finding strategy preserved the existing concrete factory sheds, adding three monumental board-marked concrete towers with irregular, unglazed window apertures that evoke cave-like openings. This tectonic approach combines technological

know-how with the spontaneity of primitive art, utilizing simple wooden row seating in the theatre to foster critical engagement.

Possible visuals: Sections through the concrete sports towers, detailed drawings of the wooden theatre seating, and photographs of the internal water pool and fireplace are highly recommended.

Source leads: Consult Lina Bo Bardi's project writings, monographs from the Instituto Bardi, and reviews in the Architectural Review.

Case study name: Teshima Art Museum

Architect or practice: The project was designed by Ryue Nishizawa and Rei Naito on Teshima Island, Japan, in 2010.

Why it may fit: This museum belongs to the category of elemental and ethereal architecture, resembling a single water droplet resting on a restored agricultural hillside to create a column-free, open gallery. This famous project aligns with the position by using nature as its primary content, allowing wind, sounds, and rain to filter through large roof openings.

Form-finding strategy: The form-finding strategy involved pouring a seamless, twenty-five-centimeter-thick white concrete shell over a massive mound of earth, which was then excavated from beneath once the concrete had cured. This approach creates a low, organic interior where water continuously springs from the ground, encouraging quiet, sensory contemplation.

Possible visuals: Photographs of the two elliptical ceiling openings, detailed structural plans of the concrete shell, and images of water droplets sliding across the resin-treated floor are recommended.

Source leads: Review project profiles from the Benesse Art Site Naoshima and monographs on Ryue Nishizawa's independent practice.

Case study name: Blur Building

Architect or practice: The project was designed by Diller Scofidio in Yverdon-les-Bains, Switzerland, in 2002.

Why it may fit: This temporary pavilion belongs to the category of computational or experimental form-finding, challenging the visual bias of architecture by using water mist as its primary building material. This famous project aligns with the position by using a "white-out" to dampen sight, forcing visitors to navigate using their somatic and acoustic senses.

Form-finding strategy: The form-finding strategy utilized a lightweight, steel tensegrity frame equipped with over thirty-one thousand high-pressure water nozzles controlled by an automated computer system that responds to local wind, humidity, and temperature. This approach creates an ever-changing, cloud-like cloud over Lake Neuchâtel, proving that technology can be used to generate sensory experiences rather than sterile structures.

Possible visuals: System schematics of the automated hydraulic nozzles, photographs of the cloud-like exterior, and diagrams of the pedestrian circulation bridges are highly useful.

Source leads: Read Elizabeth Diller and Ricardo Scofidio's book, *Blur: The Making of Nothing*, and essays in *Postmodern Culture*.

Case study name: Igualada Cemetery

Architect or practice: The project was designed by Enric Miralles and Carme Pinós in Barcelona, Spain, in 1994.

Why it may fit: This landscape cemetery belongs to the category of liminal ritual spaces, conceived as a processional earthwork that brings the bereaved down into a sunken "city of the dead" to ease the tensions of grief. This famous project aligns with the position by holding the

contradictions of life and death, using raw, weathered materials to ground the body in the passing of time.

Form-finding strategy: The form-finding strategy carved a winding pathway into the quarry-like terrain, lining the route with repeatable concrete burial vaults that double as retaining walls and embedding old timber railway sleepers into the concrete path. This tectonic approach adapts the cemetery to its site, creating an in-between place where the dead and the living are brought closer together.

Possible visuals: Longitudinal sections showing the descent into the earth, detailed drawings of the embedded timber sleepers, and atmospheric photographs of the concrete burial walls are recommended.

Source leads: Refer to Enric Miralles's project essays and reviews of Catalan architecture from the late twentieth century.

Case study name: Ningbo History Museum

Architect or practice: The project was designed by Wang Shu and Lu Wenyu in Ningbo, Zhejiang Province, China, in 2008.

Why it may fit: This massive museum belongs to the category of beauty through process, acting as a tectonic mountain of memory built from the debris of demolished local villages. This famous project aligns with the position by holding the contradiction of historical trauma and future progression, using tactile materials to establish a physical connection with the region's heritage.

Form-finding strategy: The form-finding strategy combined modern reinforced concrete with the traditional "wa pan" technique, where local craftsmen hand-laid millions of recycled bricks and roof tiles dating back hundreds of years on the lower half of the facade. This approach is an rejection of China's rapid demolition and renewal projects, preserving memories of the people who lived in the area.

Possible visuals: Detailed elevation drawings of the mosaic-like brickwork, photographs of the bamboo-formed concrete walls, and plans illustrating the internal courtyards are highly effective.

Source leads: Read Wang Shu's Pritzker Prize acceptance essay and interviews published in Domus and Dezeen.

Case study name: Bruder Klaus Field Chapel

Architect or practice: The project was completed by Peter Zumthor in Mechernich-Wachendorf, Germany, in 2007.

Why it may fit: This rural chapel is a famous masterpiece of embodied phenomenology, using darkness, texture, and scent to create an intense, inward-focused sanctuary. It aligns with the position by rejecting conventional religious symbols, relying entirely on the physical traces of its construction to prompt profound emotional reflection.

Form-finding strategy: The form-finding strategy involved ramming concrete by hand in twenty-four layers over an inward-leaning tepee structure composed of one hundred and twelve spruce logs, which were then slowly burned out from within to leave a charred, textured interior cavity. This process preserves the negative cast of the timber, allowing daylight to filter through a narrow oculus onto the hand-poured lead-and-tin floor.

Possible visuals: Diagrams of the wooden tepee formwork, photographs of the hand-poured lead-and-tin floor, and images of light streaming through the star-shaped glass bulbs are essential.

Source leads: Consult James Pallister's Sacred Spaces, and monographs dedicated to Zumthor's material practice.

Case study name: Soft and Hairy House

Architect or practice: The project was completed by Ushida Findlay Architects in Tsukuba, Japan, in 1994.

Why it may fit: Inspired by Salvador Dali's surrealist writings, this organic home is a deep cut in the category of subtly surreal dreamscapes, exploring the continuity between human bodies and domestic space. It aligns with the position by using playful, curvaceous forms to challenge the rigid modernism of suburban Japan, framing the home as a psychological projection of its inhabitants.

Form-finding strategy: The form-finding strategy coiled the house around a central courtyard and finished it in a continuous texture of brush-finished mortar, with a roof covered in a wild carpet of native grasses. This tectonic approach blends the rough simplicity of "wabi-sabi" with organic geometry to stimulate multisensorial perception.

Possible visuals: Surrealist-inspired process sketches, plans demonstrating the coiled spatial layout, and photographs of the overgrown grass roof are highly recommended.

Source leads: Review articles by Mark Titman in international journals and early portfolios of the Ushida Findlay partnership.

Case study name: Echo Chamber

Architect or practice: The project was completed by Ushida Findlay Architects in Tokyo, Japan, in 1989.

Why it may fit: This early residential project is a deep cut in tectonic whimsy, exploring body-space continuity and non-rectilinear, neo-expressionist design. It aligns with the position by prioritizing the psychological desires of its inhabitants, utilizing unusual topological forms to activate sensory perception.

Form-finding strategy: The form-finding strategy utilized the mathematics of topology and the geometry of chaos to generate a flowing concrete envelope that rejects standard modernist angles. This approach creates a continuous, expressive surface that wraps around the living areas, challenging conventional boundaries between interior and exterior.

Possible visuals: Topological geometry diagrams, floor plans showing the organic circulation loops, and photographs of the textured concrete surfaces are recommended.

Source leads: Consult ArchiLab's catalog archives and historical publications on the Ushida Findlay partnership.

Case study name: Quincho Tia Coral

Architect or practice: The project was completed by Gabinete de Arquitectura in Asunción, Paraguay, in 2013.

Why it may fit: This private dining pavilion is a deep cut in the category of tectonic whimsy, utilizing low-cost local brick to create a playful, shaded outdoor space. It aligns with the position by showing how structural experimentation can turn common, inexpensive materials into a highly expressive, beautiful canopy.

Form-finding strategy: The form-finding strategy involved prefabricating brick-and-mortar panels on the ground and tilting them into non-standard, self-supporting screen walls and vaults that filter daylight. This tectonic approach explores the properties of brick under compression, creating an intimate, shaded sanctuary that engages the senses.

Possible visuals: Construction photos showing the prefabrication of the brick panels, structural sections of the vaults, and photographs of the filtered light patterns are highly useful.

Source leads: Refer to Nick Gelpi's book, *The Architecture of Full-Scale Mock-Ups*, and interviews with Solano Benítez.

Insert table here:

- Precedent Count: 21 total case studies longlisted.
- Era Representation: Mid-century, Late Twentieth Century, and Early Twenty-First Century.
- Geographical Range: Japan, Europe, South America, West Africa, and the Middle East.

Stronger Shortlist

From the comprehensive longlist, nine exceptional projects are shortlisted as the most potent precedents for the developed argument.

1. House & Restaurant by Junya Ishigami

This project is exceptionally strong because it represents a radical reversal of typical construction order, embracing unpredictability and material dirt. It directly supports the argument that beauty can emerge from chaotic, physical processes rather than preconceived digital images. Rather than presenting a generic, hyper-minimalist white box, Ishigami forces a raw, heavy, earth-bound geology to become a domestic and commercial refuge, celebrating the beauty of imperfection. This connects directly to Gaston Bachelard's theories of the primeval cave and material imagination.

- Visuals: 3D point cloud scans of the warped glass apertures and excavation site photographs.
- Risks: The extreme cost and six-year construction period make it difficult to advocate as a replicable model for everyday architecture.

2. PC Caritas by Architecten de Vylder Vinck Taillieu

This project is a powerful precedent because it uses architectural ruins to host psychological healing, challenging the sterile, clinical codes of modern healthcare. It supports the argument that spatial care must hold contradictions of memory, decay, and vulnerability to offer true dignity. This moves far beyond generic wellness design, illustrating a raw, tactile empathy that aligns with the theories of Maurice Merleau-Ponty.

- Visuals: Construction photos showing concrete block masonry insertions and drawings of the internal greenhouses.
- Risks: The open-air, roofless design may raise practical concerns regarding thermal comfort and local building code regulations.

3. Ca'n Terra by Ensamble Studio

This project is strong because it demonstrates a post-digital form-finding strategy where the architect's hand remains in the background, allowing the existing geological mass to dictate domestic life. It supports the argument that technology, such as 3D laser scanning, should be used to reveal pre-existing relationships and structure intuition rather than generate slick, empty digital complexity.

- Visuals: Monochromatic point-cloud scans of the subterranean stone voids and site plans.
- Risks: The highly specific underground microclimate may require mechanical intervention to manage humidity for long-term habitation.

4. Sancaklar Mosque by Emre Arolat Architects

Sancaklar Mosque is a powerful precedent because it strips away all cultural and historical ornamentation to focus entirely on the physical and emotional essence of space. It supports the argument that the body understands sacred space through light, shadow, and ground levels before the intellect can apply cultural meanings. This connects to the phenomenology of sacred space and the rejection of historicist pastiche.

- Visuals: Site sections showing the underground descent and photos of light casting over the fractured slate wall.
- Risks: The stark, minimalist concrete interior could be perceived as cold or uninviting by traditional worshippers.

5. Water Cathedral by GUN Architects

This temporary pavilion is highly successful because it uses water as an active structural and atmospheric building block. It supports the argument that form-finding should respond to transient elements and microclimates. This represents a highly playful and delicate form of tectonic whimsy that engages the body through cooling mist and acoustic droplets.

- Visuals: Detailed assembly drawings of the suspended fabric prisms and water flow diagrams. * Risks: As a temporary structure, its long-term durability and structural maintenance under varying wind loads are untested.

6. Teletón Children's Rehabilitation Center by Gabinete de Arquitectura

The strength of this project is its ability to extract monumental beauty and structural drama from low-cost, ordinary brick. It supports the argument that spatial care must be accompanied by material warmth and tectonic innovation to empower vulnerable occupants. This project connects directly to Kenneth Frampton's theories of Critical Regionalism.

- Visuals: Photos of the brick-and-mortar trusses and plans of the shaded circulation paths.
- Risks: The manual fabrication of the custom brick panels requires highly skilled on-site coordination and physical labour.

7. Lamune Onsen by Terunobu Fujimori

This project is strong because it blends ancient vernacular craft with a highly expressive, almost surreal dreamscape aesthetic. It supports the argument that tectonic whimsy is a legitimate form of architectural intellect, engaging the body through carbonated waters and charred textures. It directly illustrates the Japanese philosophy of "wabi-sabi" or imperfect perfection.

- Visuals: Elevation details of the black-and-white striped cedar walls and photos of the living pine roof towers.
- Risks: Sprouting live trees from copper towers poses ongoing risks of roof leakage and biological rot.

8. Bruder Klaus Field Chapel by Peter Zumthor

The ultimate expression of material phenomenology, this chapel is strong because its form is a direct register of its raw construction history. It supports the argument that beauty emerges through process and restraint rather than visual image-making. This connects to Gaston Bachelard's concept of the "material imagination".

- Visuals: Drawings of the wooden tepee formwork and photos of the charred, textured concrete interior.
- Risks: The extremely tight interior can only accommodate a few visitors at a time, limiting its capacity as a public gathering space.

9. Blur Building by Diller Scofidio

This project is exceptionally strong because it challenges the visual bias of architecture, using water mist as its primary cladding material. It supports the argument that technology and automated computational systems can be used to generate sensory experiences rather than sterile, empty structures. This connects directly to the post-digital trend of "anexact architecture".

- Visuals: System schematics of the automated hydraulic nozzles, photographs of the cloud-like exterior, and diagrams of the pedestrian circulation bridges.
- Risks: The extreme reliance on high-pressure water nozzles and nuclear engineering expertise limits its applicability to permanent, low-tech buildings.

Insert table here:

- Shortlist selection criteria: High tactile presence, non-standard structural systems, direct alignment with radical care, and clear phenomenological focus.
- Shortlisted candidates: 9 highly vetted global precedents.

Suggested Pairings or Groupings

To build a compelling, multi-dimensional architectural argument in the final 600–800 word paper, three distinct precedent pairings are suggested.

Grouping 1: The Raw Soil and the Carved Cave (Material Inversion and Process-Based Beauty)

- Case studies: House & Restaurant by Junya Ishigami and Ca'n Terra by Ensamble Studio.
- Argument: This pairing allows the designer to argue that beauty is not an preconceived digital image, but an emergent property discovered by interacting directly with geological mass. By highlighting the reversal of standard construction, the paper can demonstrate how digital technology (like 3D scans) should be used to reveal the hidden complexities of earth and stone, rather than flattening them.
- Contrast: Ishigami pours concrete into manual soil excavations to create a new, artificial cave, while Ensamble Studio uses digital coordinates to carve selective cuts into a pre-existing nineteenth-century industrial quarry.
- Risk: The argument could become overly technical or process-focused, neglecting the social or emotional experience of the human occupants.

Grouping 2: Therapeutic Landscapes and Structural Contradiction

(Radical Care and Non-Standard Tectonics)

- Case studies: PC Caritas by Architecten de Vylder Vinck Taillieu and Teletón Children's Rehabilitation Center by Gabinete de Arquitectura.
- Argument: This pairing argues that architectural care must reject the sterile, clinical, and generic standards of modern wellness architecture. Instead, true psychological safety and dignity are fostered through warm, raw, unpolished materials (like recycled brick and exposed ruins) that hold the contradictions of human vulnerability.
- Contrast: PC Caritas operates within a decaying, open-air nineteenth-century psychiatric villa, using concrete blocks to repair broken masonry. In contrast, the Teletón Center is a newly constructed, highly sophisticated series of folded brick trusses fabricated on the ground.
- Risk: Critics may argue that the unpolished, open-air, or highly tactile surfaces do not meet standard clinical hygiene and thermal regulations.

Grouping 3: Elemental Transience and Sunken Silence (Ethereal Atmosphere and Embodied Phenomenology)

- Case studies: Blur Building by Diller Scofidio and Sancaklar Mosque by Emre Arolat Architects.
- Argument: This pairing explores how space can engage the human body before the intellect can formulate rational explanations. It argues that sequencing atmospheres—either through the cooling transience of water droplets or the heavy silence of subterranean stone—structures human movement, memory, and ritual.
- Contrast: Blur Building is a lightweight, temporary tensegrity structure that uses automated computation to manage thirty-one thousand nozzles. Sancaklar Mosque is a permanent, underground stone sanctuary that relies on silent daylight and earth mass.
- Risk: The temporary nature of a public pavilion might feel disconnected from the permanent, civic character of a mosque.

Relevant Theories, Authors, and Trends

To ground the developed argument in established scholarship, several key architectural theories and authors are presented.

Gaston Bachelard and the Material Imagination (Essential)

Bachelard's work on the psychological resonance of home, shelter, and the elements is crucial for validating the subtly surreal dreamscapes category. His exploration of how physical spaces—such as the nest, the cellar, and the cave—connect to human memory and daydreaming directly supports the cave-like, subterranean form-finding strategies of Ishigami and Ensemble Studio.

Juhani Pallasmaa and "The Eyes of the Skin" (Essential)

Pallasmaa's critique of the visual bias in modern architecture is essential for framing the embodied phenomenology category. His arguments regarding how materials, textures, and

shadows engage the human body before the intellect can process them directly support the works of Peter Zumthor, Sigurd Lewerentz, and Emre Arolat.

Kenneth Frampton and Critical Regionalism (Essential)

Frampton's framework explains how local craft, regional materials, and climate-responsive tectonics can resist generic, globalised consumerism. This theory directly supports Solano Benítez's Paraguay brick structures and Francis Kéré's Gando primary school.

The Japanese Philosophy of Wabi-Sabi in Modern Tectonics (Useful)

This traditional aesthetic concept, centering on the beauty of the imperfect, impermanent, and incomplete, is essential for framing the works of Terunobu Fujimori and Kathryn Findlay. It provides an academic vocabulary to explain how rough mortar, charred cedar, and weathering do not diminish architectural quality but enhance its emotional and tactile depth.

Topological Mathematics and Chaos Theory (Useful)

This scientific trend explores how non-linear mathematics, the geometry of chaos, and logarithmic curves can generate fluid, organic forms. It is highly useful for explaining the early, non-rectilinear works of Ushida Findlay.

Bertolt Brecht's Verfremdungseffekt (Distancing Effect) in Space (Optional)

Lina Bo Bardi explicitly used Brechtian theory to design the seating and materials of the SESC Pompéia theatre. This theory explains how raw, unpainted cinder blocks and hard wooden benches prevent sensory passivity, forcing users to remain active, critical, and engaged protagonists in the space.

Parametricism (Probably Unnecessary)

While parametric computation is technically sophisticated, its focus on slick, digital complexity lacks the tectonic whimsy, material empathy, and phenomenological grounding central to the developed position. It should be avoided.

Insert table here:

- Theoretical framework priority: High priority given to material imagination, wabi-sabi, and critical regionalism.
- Aesthetic alignment: Theoretical models must defend raw tactile experience over sterile visual perfection.

Possible Visuals

To enrich the developed architectural argument, a carefully curated selection of drawings, diagrams, and photographs should be integrated.

First, point-cloud scans and digital surveys of Ca'n Terra should be used to illustrate how

advanced technology can capture geological variations and guide minimal, intuitive cuts.

Second, construction process photographs of La Trufa , showing the hay-bale armature, the concrete pour, and the calf Paulina eating the formwork, will visually demonstrate a metabolic, process-driven approach to beauty.

Third, detailed structural sections of PC Caritas must be included to reveal how concrete-knit brickwork and internal greenhouses can be woven into a ruined shell to construct a landscape of radical care.

Fourth, technical construction details of Sigurd Lewerentz's Malmö Eastern Cemetery Flower Kiosk —specifically showing the frameless glass windows clipped directly to the raw, uneven brickwork—will illustrate tectonic whimsy and material honesty.

Finally, atmospheric photographs of Sancaklar Mosque , depicting daylight filtering through the fractured stone Qiblah wall, will visually reinforce how spatial sequence and natural light can ground the human body.

Suggested 600–800 Word Argument Structure

The final assignment section can be structured using the following logical sequence:

First, introduce the core architectural position. Frame beauty as the primary vector of spatial attraction, asserting that spatial intelligence and embodied phenomenology ground the body before the intellect can rationalise the space.

Second, integrate the AI dialogue reflection. Explain how the computational process was used to structure raw, intuitive design impulses without flattening human emotion. Briefly use Junya Ishigami as a calibration reference, showing how his work bridges lightness with heavy, process-driven earthworks.

Third, present the first case study (e.g., PC Caritas). Focus on its form-finding strategy of working directly with ruined brickwork, showing how it embodies radical care, emotional safety, and structural contradiction. Connect this to Merleau-Ponty's phenomenology.

Fourth, present the second case study (e.g., Ca'n Terra). Explain how the digital scanning of geological mass represents a post-digital form-finding strategy that prioritises the beauty of the earth over imposed geometric forms. Connect this to Gaston Bachelard's material imagination.

Fifth, synthesise the two precedents. Contrast the open-air, repaired brick ruin of PC Caritas with the carved, stone cave of Ca'n Terra. Argue that both projects demonstrate how turning spatial chaos into beautiful, liveable structure requires radical empathy and an acceptance of material imperfection.

Finally, close the section with a transition to the upcoming conclusion, preparing to reflect on how these precedents expand the established design language.

Support for Later Conclusion

This research provides robust evidence to help the designer construct a 100–200 word conclusion regarding whether the AI dialogue shifted or reinforced their architectural thinking.

First, the research reinforces the designer's core position that beauty is an essential, primary force, as validated by the rigorous phenomenological craft of Sigurd Lewerentz and Peter Zumthor.

Second, the research complicates the designer's thinking by proving that digital technology and advanced computation (such as Ensamble Studio's 3D point-cloud scans or Ishigami's iPad-guided excavations) do not have to result in slick, generic, or sterile spaces. Instead, these tools can be used to precisely capture, protect, and celebrate the beautiful, unpredictable irregularities of physical matter.

Third, the research shifts the designer's perspective on "radical care" and "refuge." It demonstrates that psychological safety does not require soft, generic "wellness" curves; rather, it can be powerfully achieved through raw, cracked, and heavily textured environments (such as PC Caritas) that reflect and respect the complex, vulnerable nature of the human condition.